

miniLISA: Design and Component Noise Model of an Optical Time-Delay Interferometry Testbed

Author list to be confirmed
Max Planck Institute for Gravitational Physics
(Dated: October 2, 2026)

miniLISA is a laboratory testbed for testing laser- and clock-noise suppression together with gravitational-wave signal recovery. Optical inputs provide laser phase noise and modulation sidebands, while FPGA delay lines represent inter-spacecraft propagation and add prescribed signal responses. We relate the synthesized readouts to a reduced LISA measurement model and propagate measured electronic and modulation spectra through second-generation time-delay interferometry with clock correction. An order-of-magnitude detector calculation lies below these contributions for the assumed optical powers. The electronic baseline is the larger measured contribution over most of the plotted band. The common board-reference fluctuation is suppressed by TDI and clock correction; differential board timing is assumed negligible in the component sum. This assumption and the remaining delay errors require experimental verification.

I. INTRODUCTION

Gravitational waves provide a unique observational window into astrophysical and cosmological phenomena. The first direct detection in 2015 established gravitational-wave observations of compact binary mergers with ground-based interferometers [1]. Accessing gravitational waves at millihertz frequencies requires a space-based detector to overcome terrestrial seismic and Newtonian noise and to provide sufficiently long interferometric baselines. The Laser Interferometer Space Antenna (LISA), a space-based observatory led by ESA with NASA participation, is designed to operate in the 0.1 mHz to 1 Hz band [2].

The LISA constellation consists of three spacecraft arranged in an approximately equilateral triangle with arm lengths of about 2.5 million km. Laser beams exchanged between the spacecraft form a constellation-scale interferometer. Unlike ground-based detectors with effectively equal arms, the LISA arm lengths vary by up to about 1.2% due to orbital motion, preventing direct cancellation of laser frequency noise. Although cavity pre-stabilisation reduces laser frequency noise substantially, it remains several orders of magnitude above the displacement noise budget required for gravitational-wave detection [3]. To suppress this noise, time delay interferometry (TDI) has been developed as a post-processing technique that synthesizes virtual equal-arm interferometers by appropriately time-shifting and combining phasemeter measurements.

In addition to laser frequency noise, LISA measurements are affected by clock noise originating from the independent ultra-stable oscillators (USOs) carried by each spacecraft. Since all phase measurements are referenced to these clocks, timing jitter couples directly into the measured heterodyne beatnotes. Space-qualified USOs achieve Allan deviations of order 10^{-13} at one-second averaging times [4], which is insufficient for the picometer-level test-mass position sensitivity required by LISA. To mitigate this effect, LISA employs additional clock-

noise transfer measurements—implemented via modulation sidebands exchanged between spacecraft.

TDI has been developed analytically and tested through numerical simulations [3, 5]. Simulations predict the response of a specified measurement model and test its numerical implementation. Physical experiments provide a complementary test of whether the hardware follows that model, including couplings and processing effects that an idealized simulation may omit.

The University of Florida Laser Interferometry Simulator (UFLIS) demonstrated TDI and TDI-ranging with time-varying electronic propagation delays and optical laser signals [6]. miniLISA continues this approach with FPGA delay lines and independent spacecraft clock references. The experiment of de Vine et al. demonstrated laser-noise suppression of approximately 10^9 and clock-noise suppression of 6×10^4 , using short optical paths [7]. LISA On Table demonstrated first-generation TDI with simulated noise and static unequal arms [8].

Here, we present an overview of an ongoing effort to develop a TDI testbed that combines realistic laser frequency noise, programmable delay lines, and independent clock sources within a single experimental framework. The testbed focuses on reproducing the dominant noise couplings of the LISA metrology system—specifically, laser frequency noise and clock noise—with component-noise propagation through time-dependent delays. Crucially, the same delay lines that impose the inter-spacecraft light-travel times also allow physically consistent gravitational-wave strain signals to be injected directly into the measurement chain. The testbed can therefore be used not only to demonstrate noise suppression, but to verify that a known gravitational-wave signal survives TDI processing of hardware-generated data with the expected amplitude and phase.

An initial purely electrical implementation centered around field programmable gate array (FPGA)-based digital delay lines has already been demonstrated in Ferguson et al. [9]. The developments discussed here extend that architecture by incorporating an optical front-

end with physically representative laser noise and independently running spacecraft clocks. This reproduces the laser-noise and propagation structure of the one-way measurements together with a reduced clock-transfer model, allowing TDI combinations and clock-noise calibration to be formed entirely from hardware-generated data. The goal of this work is to realize a laboratory-scale analog of the LISA inter-satellite metrology system—a “miniLISA”—that bridges the gap between idealized software simulations and the flight instrument, and provides a controlled environment for testing the robustness of TDI to instrumental non-idealities and realistic gravitational-wave signal injection.

More concretely, we look for residual laser and clock noise below the secondary-noise floor after processing, together with recovery of an injected GW signal at the expected amplitude and phase.

Section II introduces the LISA measurements and noise-suppression combinations. Section III describes their laboratory implementation, and Section IV calculates the contributions from measured and modelled noise sources. We then discuss possible extensions and remaining measurements.

II. LISA INTERFEROMETRIC DETECTION SYSTEM

We first write the LISA phase measurements and identify how laser and clock noise enter them. These expressions provide the reference for the laboratory readouts in Section III.

A. Interferometric Phase Measurements

Each LISA spacecraft carries two optical benches, with one laser on each bench. On a given bench, the local laser is compared with the laser on the other bench and with light received from a distant spacecraft. The science interferometer measures the inter-spacecraft phase difference; the test-mass and reference interferometers provide the local measurements needed to remove optical-bench motion. The upper panel of Fig. 1 shows the science-interferometer part of a two-way link.

For the laser on movable optical system assembly (MOSA) ij , on spacecraft i facing spacecraft j , we write

$$\Phi_{ij}(t) = \nu_{ij}t + \delta\phi_{ij}(t), \quad (1)$$

where ν_{ij} denotes the nominal carrier frequency and $\delta\phi_{ij}(t)$ represents the intrinsic laser phase noise. Throughout this section, phases are expressed in cycles, frequencies in Hz, and clock errors in seconds. Capital Φ denotes total phase, while $\delta\phi$ denotes phase fluctuations about the nominal linear evolution. We use a single time coordinate, neglect relativistic time transformations, and retain constant nominal frequencies in the noise-coupling

coefficients. The clock on spacecraft i reads $t_i = t + q_i(t)$, where q_i is its timing error.

An electro-optic modulator (EOM) driven by the local USO adds phase-modulation sidebands to each laser field. Their modulation phase contains $\nu_{ij}^m q_i(t)$, so the transmitted light carries the local clock fluctuations. The clock error has one spacecraft index because both benches use the same USO.

The modulation tones are derived from the USO at approximately 2.4 GHz. A 1 MHz difference between the relevant tones separates the sideband–sideband beat notes from the carrier–carrier beat note. Excess noise in the tone-generation chain enters the clock-transfer measurement and is considered below.

The modulation depth $\mu_{\text{EOM}} = 0.53$ rad places about 87% of the optical power in the carrier and 6.5% in each first-order sideband. The USO references the modulation, sampling-clock and pilot-tone synthesis chains; their role in clock-noise suppression is discussed in Section II C. Additional phase modulation with pseudo-random codes supports ranging and data transfer; it is distinct from the sinusoidal clock-transfer modulation considered here [10].

Propagation from MOSA ij to ji delays the emitted phase and adds the GW response along the arm. Following the separation of background propagation and GW perturbations in [3], we write

$$d_{ji}^{\text{tot}}(t) = d_{ji}(t) + H_{ji}(t). \quad (2)$$

Here $d_{ji}(t)$ is the background propagation time, which can vary slowly with the constellation breathing, and $H_{ji}(t)$ is its GW-induced perturbation. The first index denotes the receiving spacecraft and the second the emitting spacecraft. We define the delay operator using only the background propagation time,

$$\mathbf{D}_{ji}x(t) = x(t - d_{ji}(t)). \quad (3)$$

Applying the total propagation time to Eq. (1) and expanding to first order gives

$$\begin{aligned} \Phi_{ij}(t - d_{ji}(t) - H_{ji}(t)) &\simeq \nu_{ij} [t - d_{ji}(t)] \\ &\quad + \mathbf{D}_{ji}\delta\phi_{ij}(t) - \nu_{ij}H_{ji}(t), \end{aligned}$$

where the product $H_{ji}\mathbf{D}_{ji}\delta\phi_{ij}$ of the GW perturbation and laser-frequency noise has been neglected. We therefore introduce the GW phase contribution

$$\delta\phi_{ji}^{\text{GW}}(t) = -\nu_{ij}H_{ji}(t). \quad (4)$$

To this order, the received total phase is $\mathbf{D}_{ji}\Phi_{ij}(t) + \delta\phi_{ji}^{\text{GW}}(t)$: the GW signal is an additive phase contribution, separate from the background delay operator. For constant ν_{ij} , the corresponding frequency fluctuation is $\dot{\delta\phi}_{ji}^{\text{GW}} = -\nu_{ij}\dot{H}_{ji}$.

At spacecraft j , the received field i interferes with the local laser field of phase $\Phi_{ji}(t)$ on a photodetector. The resulting heterodyne signal contains both the carrier–carrier beat note and the sideband–sideband beat

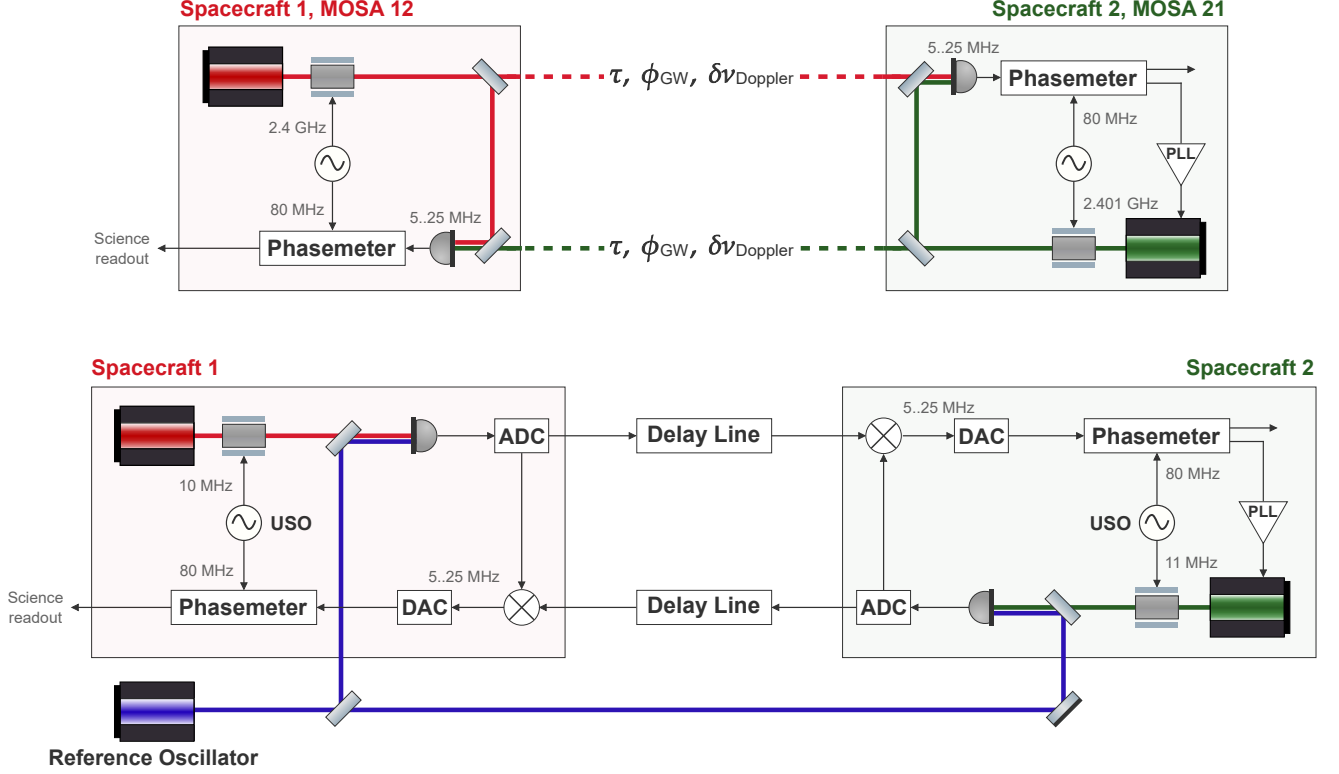


FIG. 1. Two-way science links in LISA (upper) and miniLISA (lower). In LISA, the received laser field interferes with the local field after propagation between spacecraft. Optical sidebands carry the transmitting clock fluctuations, and the receiving clock references the phase measurement. In miniLISA, a common reference laser produces MHz input beat notes, and electronic delay lines reproduce the propagation and subtraction of remote and local signals. The lower-panel frequency and clock labels require the updates noted below.

Draft note: Revise the lower-panel labels to independently referenced Mokus, (35,36,34) MHz modulation and the synthesized output tones for all six links. The legacy USO and 10/11 MHz labels do not describe the operating point used here.

notes. Additional cross terms lie outside the heterodyne range of the phasemeter; a representative LISA frequency plan uses absolute beat frequencies from 5 to 25 MHz [11].

The electrical signal from the photodetector is then digitized by the phase measurement system (PMS) front-end electronics. The sampling process introduces a coupling of the local USO timing error $q_j(t)$ into the measured phase. For all measurement readouts, we adopt the LISA convention that phases are written as “distant minus local.”¹ We define the nominal carrier coefficient

$$\alpha_{ji} = \Delta\nu_{ji} = \nu_{ij} - \nu_{ji}. \quad (5)$$

After removing the nominal beat evolution, the sampled carrier phase fluctuations are

$$\begin{aligned} \delta\phi_{ji}^c(t) = & \mathbf{D}_{ji}\delta\phi_{ij}(t) - \delta\phi_{ji}(t) \\ & + \delta\phi_{ji}^{GW}(t) - \alpha_{ji}q_j(t). \end{aligned} \quad (6)$$

¹ The phasemeter measures the absolute beat frequency; its sign is assigned using the frequency plan.

Here α_{ji} is the beat frequency in the fixed-frequency approximation used for the noise model. For an orbit-dependent frequency plan, it would be replaced by the actual signed beat frequency, including Doppler shifts.

The signal is then passed to digital phase lock loops (DPLLs), which track the carrier and both sideband beat notes. In the following readouts, the nominal phase evolution is largely removed from the tracked phase, leaving the fluctuations used for GW detection and clock-noise suppression. To connect this noise model to the measurements used for TDI, we also include the secondary disturbances in the science readout:

$$\begin{aligned} \text{sci}_{ji}^c(t) = & \mathbf{D}_{ji}\delta\phi_{ij}(t) - \delta\phi_{ji}(t) - \alpha_{ji}q_j(t) \\ & + \delta\phi_{ji}^{GW}(t) + N_{ji}^c(t) \\ & - \frac{\nu_0}{c} (\mathbf{n}_{ij} \cdot \mathbf{D}_{ji}\Delta_{ij}(t) + \mathbf{n}_{ji} \cdot \Delta_{ji}(t)), \end{aligned} \quad (7)$$

The term $N_{ji}^c(t)$ contains readout noise from optical path-length fluctuations and electronics. The vectors $\Delta_{ij}(t)$ are optical-bench displacements in metres. The

unit vector $\mathbf{n}_{ji}$ points along the incoming beam from spacecraft i to spacecraft j , and $\mathbf{n}_{ij} = -\mathbf{n}_{ji}$ in the geometry used for these displacement terms. The factor $\nu_0/c = 1/\lambda_0$ converts path-length fluctuations to phase in cycles, with $\nu_0 \simeq 281$ THz for the nominal wavelength $\lambda_0 = 1064$ nm. MHz-scale offsets between lasers are small compared with this optical frequency, so a common ν_0 suffices for the displacement terms. Those offsets remain essential in the heterodyne clock coefficients, which depend on frequency differences.

The corresponding upper sideband–sideband beat note readout can be written as

$$\begin{aligned} \text{sci}_{ji}^{\text{sb},+}(t) = & \mathbf{D}_{ji} \delta\phi_{ij}(t) - \delta\phi_{ji}(t) - \gamma_{ji} q_j(t) \\ & + \delta\phi_{ji}^{\text{GW}}(t) + N_{ji}^{\text{sb},+}(t) \\ & - \frac{\nu_0}{c} (\mathbf{n}_{ij} \cdot \mathbf{D}_{ji} \Delta_{ij}(t) + \mathbf{n}_{ji} \cdot \Delta_{ji}(t)) \\ & + \nu_{ij}^m \mathbf{D}_{ji} q_i(t) - \nu_{ji}^m q_j(t). \end{aligned} \quad (8)$$

The sideband clock coupling coefficient is

$$\gamma_{ji} = \Delta\nu_{ji} + \nu_{ji}^m - \nu_{ji}^m = \Delta\nu_{ji} + \Delta\nu_{ji}^m = \Delta\nu_{ji}^{\text{sb},+}. \quad (9)$$

Here $N_{ji}^{\text{sb},+}$ contains readout noise in the upper-sideband channel. The lower sideband has the same laser phase noise but the opposite sign for each modulation contribution. The sidebands have optical frequencies $\nu_{ij} \pm \nu_{ij}^m$, so the same path-length change produces a slightly different phase shift in each sideband. Using ν_0/c for their displacement couplings neglects a fractional correction of order $\nu^m/\nu_0 \simeq 9 \times 10^{-6}$. Similarly, the upper-sideband GW phase is $-(\nu_{ij} + \nu_{ij}^m)H_{ji}$, differing from the carrier by $-\nu_{ij}^m H_{ji}$. We neglect this small difference in the readout equations above and use the same $\delta\phi_{ji}^{\text{GW}}$ for the carrier and both sidebands. The δ denotes a perturbation to the nominal optical phase, rather than stochastic noise.

The test-mass and reference measurements are, on MOSA ji

$$\begin{aligned} \text{tm}_{ji}(t) = & \delta\phi_{jk}(t) - \delta\phi_{ji}(t) + \beta_{ji} q_j(t) + \mu_j(t) + N_{ji}^{\text{tm}}(t) \\ & + \frac{2\nu_0}{c} \mathbf{n}_{ji} \cdot (\delta_{ji}(t) - \Delta_{ji}(t)), \end{aligned} \quad (10)$$

$$\begin{aligned} \text{ref}_{ji}(t) = & \delta\phi_{jk}(t) - \delta\phi_{ji}(t) + \beta_{ji} q_j(t) \\ & + \mu_j(t) + N_{ji}^{\text{ref}}(t). \end{aligned} \quad (11)$$

The backlink term μ_j describes noise in the fibre connecting the two optical benches on spacecraft j . We assume reciprocal propagation through this fibre. The vector δ_{ji} is the test-mass displacement, including motion produced by acceleration noise.

With the same sampling convention, the reference and test-mass clock coefficient is

$$\beta_{ji} = \nu_{ji} - \nu_{jk}. \quad (12)$$

It has the opposite sign to the adjacent-minus-local beat frequency, which explains the plus sign multiplying $\beta_{ji} q_j$ in the readouts.

These equations retain the terms needed to explain laser- and clock-noise suppression. A full instrument model includes further effects, such as tilt-to-length coupling and additional stray-light and optical-path disturbances. The miniLISA model omits the test-mass and reference interferometers and their associated bench-motion, backlink, and test-mass noise terms.

B. Laser Noise and TDI

Pre-stabilisation reduces laser frequency noise before the measurements are combined. Offset phase locks keep the inter-spacecraft beat notes within the readout band, but the remaining laser noise still exceeds the required displacement sensitivity. TDI removes it by combining measurements in which the same laser fluctuation has propagated along different paths.

For scale, consider a representative pre-stabilized frequency-noise amplitude of $30 \text{ Hz}/\sqrt{\text{Hz}}$ [3]. At Fourier frequency $f = 10 \text{ mHz}$, the corresponding single-laser phase-noise amplitude is $30/(2\pi f) \simeq 480 \text{ cycles}/\sqrt{\text{Hz}}$, equivalent to about $0.5 \text{ mm}/\sqrt{\text{Hz}}$ displacement noise at 1064 nm . This illustrative conversion is many orders of magnitude above picometre-scale displacement noise required for GW detection.

The construction proceeds in two steps: local measurements first remove optical-bench motion and reduce the six laser noises to three; delayed combinations of the resulting links then suppress the remaining laser noise [12, 13]. Using Eqs. (7), (10), and (11), we form

$$\begin{aligned} T_{ji} &= \text{tm}_{ji} - \text{ref}_{ji}, \\ \xi_{ji} &= \text{sci}_{ji}^c - \frac{1}{2} (T_{ji} + \mathbf{D}_{ji} T_{ij}). \end{aligned}$$

Within the common-optical-frequency approximation above, the bench terms cancel, leaving the test-mass contribution

$$M_{ji} = -\frac{\nu_0}{c} [\mathbf{n}_{ji} \cdot \delta_{ji}(t) + \mathbf{D}_{ji} (\mathbf{n}_{ij} \cdot \delta_{ij}(t))].$$

The following equations omit readout noises. Reciprocal backlink noise cancels in the local reference difference

$$z_{ji} = \frac{1}{2} (\text{ref}_{ji} - \text{ref}_{jk}) = \delta\phi_{jk} - \delta\phi_{ji} + \beta_{ji} q_j.$$

For cyclic triples $(i, j, k) = (1, 2, 3), (2, 3, 1), (3, 1, 2)$, the intermediate variables can then be defined as

$$\begin{aligned} \eta_{ji} &= \xi_{ji} - z_{ji} \\ &= \mathbf{D}_{ji} \delta\phi_{ij}(t) - \delta\phi_{jk}(t) - (\alpha_{ji} + \beta_{ji}) q_j(t) \\ &\quad + \delta\phi_{ji}^{\text{GW}}(t) + M_{ji}(t), \\ \eta_{ij} &= \xi_{ij} + \mathbf{D}_{ij} z_{ji} \\ &= \mathbf{D}_{ij} \delta\phi_{jk}(t) - \delta\phi_{ij}(t) - \alpha_{ij} q_i(t) \\ &\quad + \mathbf{D}_{ij} [\beta_{ji} q_j(t)] \\ &\quad + \delta\phi_{ij}^{\text{GW}}(t) + M_{ij}(t). \end{aligned}$$

Here M_{ij} is obtained by exchanging i and j in M_{ji} . These combinations reduce the number of independent laser phase noises from six to three, retaining $\delta\phi_{12}$, $\delta\phi_{23}$ and $\delta\phi_{31}$, while removing the optical-bench displacement terms. The construction introduces an asymmetry between the two link orientations.

For a Michelson combination centred on spacecraft i , we first form the round-trip measurements

$$R_{ij} = \eta_{ij} + \mathcal{D}_{ij}\eta_{ji}, \quad R_{ik} = \eta_{ik} + \mathcal{D}_{ik}\eta_{ki}.$$

The first-generation combination is

$$X_i = (1 - \mathcal{D}_{ij}\mathcal{D}_{ji})R_{ik} - (1 - \mathcal{D}_{ik}\mathcal{D}_{ki})R_{ij}. \quad (13)$$

Here $\mathbf{D}$ describes propagation in the measurement chain, while $\mathcal{D}$ denotes a delay applied in post-processing. Cancellation requires the processing delays to match the physical delays. A mismatch δd leaves a phase residual proportional to $\delta d \delta\phi$, so either an inaccurate delay estimate or an untracked change in the physical delay can limit laser-noise suppression.

First-generation TDI combinations cancel laser noise for constant light-travel times. Orbital motion makes these delays time-dependent, so their order of application matters and the cancellation is no longer exact. Second-generation combinations use longer sequences of delays to cancel the leading residual from arm-length changes, leaving terms involving higher powers of the arm velocities and arm accelerations [13].

C. Clock-noise Coupling and Correction

Clock error enters a carrier phase measurement multiplied by the local beat frequency. The optical sidebands provide an additional measurement of the transmitted clock phase, allowing the remote and local clocks to be compared [14]. Subtracting the carrier measurement from the upper-sideband measurement removes their common laser phase. Neglecting readout and modulation noise for the moment, we obtain

$$r_{ji} = \frac{\text{sci}_{ji}^{\text{sb},+}(t) - \text{sci}_{ji}^{\text{c}}(t)}{\nu_{ij}^m} = \mathbf{D}_{ji}q_i(t) - q_j(t). \quad (14)$$

The retained GW phase cancels in this difference because we have approximated the carrier and sideband GW couplings as equal. Keeping their optical-frequency difference would instead add $-H_{ji}$ to r_{ji} ; the cancellation is therefore approximate. The normalization uses the transmitted modulation frequency ν_{ij}^m , giving r_{ji} units of seconds. Equivalently, $r_{ji} = (\text{sci}_{ji}^{\text{sb},+} - \text{sci}_{ji}^{\text{sb},-})/(2\nu_{ij}^m)$ under the same approximations. These variables contain delayed remote clock noise minus local clock noise, matching the structure of laser noise in the science readouts. Delayed combinations of the r variables can therefore be used to construct clock corrections for the TDI

observables [14]. The modulation tone must carry the same timing fluctuations that enter the phase measurement. In LISA, pilot-tone correction removes the ADC sampling jitter, making the pilot tone the relevant clock reference. Noise added between this reference and the modulation tone limits the subsequent clock correction. Let $\tau_{ij}^{\text{mod}}(t)$ denote differential timing noise in the tone derivation for MOSA ij , in seconds, and let $\psi_{ij}^{\text{mod}}(t)$ denote additional optical modulation phase noise, in cycles. The excess upper-sideband phase is

$$m_{ij}(t) = \nu_{ij}^m \tau_{ij}^{\text{mod}}(t) + \psi_{ij}^{\text{mod}}(t).$$

Including these terms and readout noise gives

$$r_{ji} = \mathbf{D}_{ji}q_i - q_j + \frac{\mathbf{D}_{ji}m_{ij} - m_{ji}}{\nu_{ij}^m} + \frac{N_{ji}^{\text{sb},+} - N_{ji}^{\text{c}}}{\nu_{ij}^m}.$$

Modulation and readout noise are kept separate because their transfer functions differ. A source modulation error appears in both local and delayed measurements, while a readout error is added in a particular measurement channel. Independent sources may be uncorrelated, but repeated appearances of the same source must still be combined before calculating its output noise power.

Dividing the modulation phase by its frequency gives $m_{ij}/\nu_{ij}^m = \tau_{ij}^{\text{mod}} + \psi_{ij}^{\text{mod}}/\nu_{ij}^m$. A higher modulation frequency therefore reduces the effect of a fixed phase noise, but leaves a fixed timing noise unchanged. Improving that timing-noise floor requires a quieter tone-generation chain. When the local and transmitted frequencies differ, the local timing contribution retains the factor ν_{ji}^m/ν_{ij}^m . A measurement at one modulation frequency cannot distinguish the two contributions.

The two modulation-frequency synthesis chains can add different excess noises. Local reference-interferometer sideband measurements provide additional observables for correcting the accessible differential part [14]. This contribution is absent from the present one-modulator-per-spacecraft testbed.

III. MINILISA DESIGN

A. Experimental Concept

The optical extension combines physical laser noise, sideband clock transfer and electronic delays of several seconds. The delays place samples acquired at different times in the same measurement, as inter-spacecraft propagation does in LISA. The following model specifies the couplings that the hardware must reproduce.

B. Synthesized LISA Measurements

The design has six directed links between three spacecraft. Each spacecraft is represented by one effective laser phase $\delta\phi_i$ and one clock timing error q_i . With the first link index denoting the receiver, the target carrier measurement is

$$\delta\phi_{ij}^c(t) = \mathbf{D}_{ij}\delta\phi_j(t) - \delta\phi_i(t) + \delta\phi_{ij}^{\text{GW}}(t) - \alpha_{ij}q_i(t),$$

and the upper-sideband fluctuations are

$$\begin{aligned} \delta\phi_{ij}^{\text{sb},+}(t) = & \mathbf{D}_{ij}\delta\phi_j(t) - \delta\phi_i(t) + \delta\phi_{ij}^{\text{GW}}(t) \\ & - (\alpha_{ij} + \nu_j^m)q_i(t) + \nu_j^m\mathbf{D}_{ij}q_j(t). \end{aligned}$$

Here α_{ij} is the signed nominal carrier beat frequency and ν_i^m is the modulation frequency on spacecraft i . The sideband sampling coefficient is $\gamma_{ij} = \alpha_{ij} + \nu_j^m - \nu_i^m$; combining it with the local modulation phase gives the clock term above. Both channels use the common GW phase approximation of Section II. We call a measurement *LISA-like* when it reproduces these couplings. Modulation, board, and readout noises are added below.

The electronic delays reproduce propagation between spacecraft and add prescribed Doppler and GW frequency shifts. Heterodyning against a common reference laser brings the optical signals into the MHz input band. The reference phase becomes part of each effective laser phase and enters the same TDI cancellation.

Unlike LISA, miniLISA uses one laser per spacecraft and has no test-mass or reference interferometers. Bench-motion corrections T_{ji} and ξ_{ji} and reference differences z_{ji} are therefore unnecessary. The synthesized links reproduce the three-laser η measurements, with the reference-interferometer clock terms proportional to β_{ji} absent. Test-mass acceleration, bench motion, backlink noise and tilt-to-length coupling are omitted.

C. Optical Input

Three lasers represent the spacecraft, and a fourth laser supplies their common optical reference. Heterodyning each spacecraft laser with the reference produces a MHz beat note that retains their relative phase fluctuations. These beat notes form the inputs to the electronic delay lines.

One spacecraft laser is stabilised to a 25 cm ultra-low expansion (ULE) cavity; the reference laser is locked to a separate 30 cm cavity by the Pound–Drever–Hall method. Both cavities are housed in a vacuum chamber with three layers of aluminium thermal shielding. The target laser frequency noise is approximately $30 \text{ Hz}/\sqrt{\text{Hz}}$ in the millihertz band. The other two spacecraft lasers are phase-locked to synthesized beat notes.

Each spacecraft uses an independent timing reference for its Moku. The Moku supplies both the phase readout and the modulation tone driving the fiber-coupled

EOM. We use q_i for the timing fluctuations common to the EOM drive and the phasemeter's sampling reference. Excess noise between these paths enters the modulation and readout terms separately.

The reference laser is unmodulated, so its carrier beats with the spacecraft carrier and both sidebands. The optical frequency plan must keep these three input tones within the delay-line bandwidth. Digital subtraction then forms the carrier–carrier and sideband–sideband measurements of the target model. A possible extension with a modulated reference is discussed in Section V A.

The lower panel of Fig. 1 shows one two-way laboratory link. Taking the photodetector beat phase as spacecraft laser minus reference laser, the carrier phase before detector noise is

$$\Phi_{\text{PD},i}^c(t) = \nu_{\text{R},i}t + \delta\phi_i^{\text{eff}}(t),$$

where $\nu_{\text{R},i}$ is the signed nominal beat note frequency between the reference laser and the spacecraft (SC) i laser, and $\delta\phi_i^{\text{eff}} = \delta\phi_i - \delta\phi_{\text{R}}$ includes their relative laser phase noise. The upper-sideband phase is

$$\begin{aligned} \Phi_{\text{PD},i}^{\text{sb},+}(t) = & (\nu_{\text{R},i} + \nu_i^m)t \\ & + \delta\phi_i^{\text{eff}}(t) + \nu_i^mq_i(t) \\ & + \nu_i^m\tau_i^{\text{mod}}(t) + \psi_i^{\text{mod}}(t). \end{aligned}$$

Here, τ_i^{mod} is timing noise added by the modulation path relative to the common reference q_i , while ψ_i^{mod} is additional modulation phase noise. Noise added only by the sampling path is included in the readout terms below. These are the single-index counterparts of τ_{ij}^{mod} and ψ_{ij}^{mod} in Section II C, with the same dependence on modulation frequency. We write their combined upper-sideband phase noise as

$$m_i(t) = \nu_i^m\tau_i^{\text{mod}}(t) + \psi_i^{\text{mod}}(t).$$

The lower sideband carries the opposite modulation phase, $-\nu_i^mq_i - m_i$. If its signed electrical frequency $\nu_{\text{R},i} - \nu_i^m$ is negative, representing it at positive frequency also reverses the measured phase sign; the clock correction below uses the upper sideband.

Each photodetector output is split between the three delay boards. A board uses the local and remote inputs to synthesize the links received at its spacecraft. Figure 2 shows the six-link routing.

Photocurrent noise produces phase errors when the delay-line PLLs track the input beats. Copies of a detector signal can share this error if the tracking responses match; ADC and electronics noise added after the split also contributes to each path. Section IV A estimates the detector scale, and Section IV C describes the propagation of shared and independent sources.

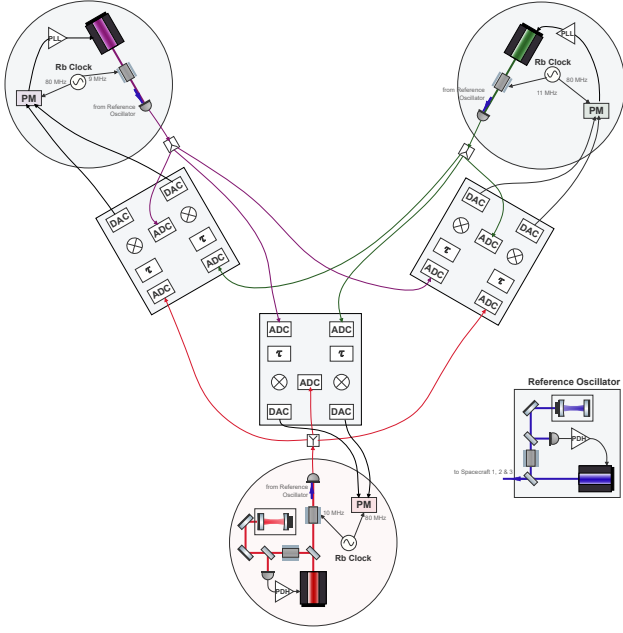


FIG. 2. Planned six-link routing architecture for miniLISA. Each spacecraft has an optical source and an independently referenced Moku phasemeter that also supplies its EOM drive; each delay-line board can synthesize two incoming links. The complete routing supports six directed links, whereas the Michelson calculation in Section IV uses links 12, 21, 13 and 31. The spacecraft-1 and reference lasers are stabilized to separate optical cavities; the other spacecraft lasers are phase-locked to synthesized beat notes. The common reference laser supplies the optical heterodyne reference.

Draft note: Update the schematic labels from the legacy modulation frequencies to (35, 36, 34) MHz and identify the independent spacecraft clock references.

D. The Delay Line

The digital delay line emulates the light-travel time along each directed link. The current firmware always tracks and stores the carrier and both sidebands. At the resulting data rate, the available memory supports a maximum delay of approximately 8.5 s, covering LISA's nominal one-way light-travel time of about 8.3 s.

The delay line uses an AMD Xilinx ZCU208 RFSoc board with integrated ADCs, DACs, FPGA logic and an embedded processing system (PS). For each link, separate DPLLs track the carrier and sidebands in the remote and local inputs. The remote frequency records pass through a buffer before subtraction of the current local frequencies. The 125 MHz processing rate limits the input tones to below 62.5 MHz; the digitally synthesized output frequencies are set separately.

Orbital delays and GW link responses are computed in advance in Python and uploaded to the PS. During playback, the PS converts the model inputs into fixed-point commands and streams timestamped records to the pro-

grammable logic. The buffer supplies the macroscopic light-travel time, while fine delay actuation follows the prescribed arm-length evolution using fractional-delay interpolation. The underlying electronic architecture is described in [9].

The FPGA subtracts the local frequency from the delayed remote frequency and adds the prescribed Doppler and GW offsets. Section III E describes the GW scaling. Omitting the nominal frequency evolution, the intended carrier phase is

$$\mathbf{D}_{ij}\delta\phi_j^{\text{eff}} - \delta\phi_i^{\text{eff}} + \delta\phi_{ij}^{\text{GW}}.$$

Section IV C propagates the component noises through ideal phase delays with changing arm lengths.

Board i receives the detector signals representing spacecraft i and j . It delays the j signal and subtracts the current i signal. The ADCs sampling these inputs and the DAC producing the synthesized beat share the board timing fluctuation ϵ_i . Their carrier-phase contributions are

$$\begin{aligned}\delta\phi_{\epsilon,ij}^c(t) &= -\nu_{R,j}\mathbf{D}_{ij}\epsilon_i(t) \\ &\quad + \nu_{R,i}\epsilon_i(t) + \alpha_{ij}\epsilon_i(t) \\ &= \nu_{R,j}(1 - \mathbf{D}_{ij})\epsilon_i(t).\end{aligned}$$

The three terms are from sampling the stored j input, sampling the current i input, and generating the output, respectively. Using $\alpha_{ij} = \nu_{R,j} - \nu_{R,i}$ gives the final delayed difference.

The same cancellation applies to the upper sideband, with the remote input frequency replaced by $\nu_{R,j} + \nu_j^m$:

$$\delta\phi_{\epsilon,ij}^{\text{sb},+}(t) = (\nu_{R,j} + \nu_j^m)(1 - \mathbf{D}_{ij})\epsilon_i(t).$$

Both expressions assume constant nominal frequencies and no additional output frequency offset.

Board timing also affects the duration of the stored delay. Let t_{in} be the physical time when the sample emerging at time t entered the buffer. A delay command d_{ij} is counted in board time $u_i(t) = t + \epsilon_i(t)$, so $u_i(t) - u_i(t_{\text{in}}) = d_{ij}$. To first order,

$$d_{ij}^{\text{phys}}(t) = t - t_{\text{in}} = d_{ij} + \delta d_{ij}^c(t), \quad (15)$$

$$\delta d_{ij}^c(t) = -(1 - \mathbf{D}_{ij})\epsilon_i(t). \quad (16)$$

This matters when the delay used in TDI differs from the actual buffer delay: the laser phases are then combined at the wrong times and do not fully cancel. The component budget assumes matched delays and excludes this residual.

E. GW Signal Injection

The GW phase defined in Eq. (4) is implemented electronically after the background delay. The required phase on the link from j to i is

$$\delta\phi_{ij}^{\text{GW}}(t) = -\nu_j H_{ij}(t).$$

Since the nominal optical frequency ν_j is constant, the frequency offset supplied to the phase accumulator is

$$\delta\nu_{ij}^{\text{inj}}(t) = \delta\dot{\phi}_{ij}^{\text{GW}}(t) = -\nu_j \dot{H}_{ij}(t). \quad (17)$$

Integrating this command gives the required GW phase, up to an initial constant. The conversion uses the represented optical frequency, rather than the MHz input beat frequency. The signal is added to the reconstructed phase after the background delay; the buffer delay itself is not changed by H_{ij} .

Each directed-link waveform is evaluated at its reception time t . The propagation and source geometry are already included in H_{ij} , so no additional background delay is applied to the injection waveform after it is generated. Relative timing between the playback streams must reproduce this convention.

The same GW offset is applied to the carrier and both sidebands, following the approximation in Section II. It therefore cancels from their difference used for clock correction while remaining in the science signal.

When the link response is supplied as a fractional-frequency signal $y_{ij} = -\dot{H}_{ij}$, the command is $\delta\nu_{ij}^{\text{inj}} = \nu_j y_{ij}$, using $\nu_j \simeq c/(1064\text{ nm})$ for the optical scale. The link responses can be computed in Python for a chosen waveform and constellation geometry. Multiple sources can be combined by summing their link responses in the linear regime. The source class is not fixed by the hardware, but the resulting commands must lie within its frequency range, resolution and update bandwidth.

A signed 32-bit GW command is added on the 64-bit phase-increment scale. At a 125 MHz accumulator clock, one least significant bit corresponds to $125\text{ MHz}/2^{64} \simeq 6.78 \times 10^{-12}\text{ Hz}$.

F. Readout and Data Processing

The electrical signals produced by the delay line are digitized and tracked by Moku phasemeters, with each spacecraft's phase measurements referenced to its own clock.

The synthesized links can be used directly as η observables. Adding in the modulation, board, and readout noises, the carrier measurement is

$$\eta_{ij}(t) = \mathbf{D}_{ij} \delta\phi_j^{\text{eff}}(t) - \delta\phi_i^{\text{eff}}(t) + \delta\phi_{ij}^{\text{GW}}(t) - \alpha_{ij} q_i(t) + \nu_{R,j}(1 - \mathbf{D}_{ij})\epsilon_i(t) + b_{ij}^c(t),$$

and the corresponding upper-sideband–upper-sideband beat note as

$$\begin{aligned} \eta_{ij}^{\text{sb},+}(t) &= \mathbf{D}_{ij} \delta\phi_j^{\text{eff}}(t) - \delta\phi_i^{\text{eff}}(t) + \delta\phi_{ij}^{\text{GW}}(t) - \gamma_{ij} q_i(t) \\ &\quad + \nu_j^m \mathbf{D}_{ij} q_j(t) - \nu_i^m q_i(t) \\ &\quad + \mathbf{D}_{ij} [\nu_j^m \tau_j^{\text{mod}}(t) + \psi_j^{\text{mod}}(t)] \\ &\quad - \nu_i^m \tau_i^{\text{mod}}(t) - \psi_i^{\text{mod}}(t) \\ &\quad + (\nu_{R,j} + \nu_j^m)(1 - \mathbf{D}_{ij})\epsilon_i(t) + b_{ij}^{\text{sb},+}(t). \end{aligned}$$

The terms proportional to q_i describe timing common to the modulation and phasemeter reference, while those proportional to ϵ_i come from the delay-line board.

The additive phase noise b_{ij}^a combines optical readout, residual board, and final-phasemeter noise, in cycles; timing and modulation noise are included explicitly. To retain the noise shared by copies of a photodetector signal, write

$$b_{ij}^a = \mathbf{D}_{ij} p_j^a - p_i^a + e_{ij}^a, \quad (18)$$

where p_i^a is the equivalent input phase noise in channel a before the split, and e_{ij}^a contains noise added by the individual electronic paths. This expression assumes matched tracking responses and demodulation phases, so that shared photocurrent noise gives the same phase error in each copy. Any mismatch must be included in the transfer functions rather than treating copies of p_i^a as independent sources.

The effective laser phases $\delta\phi_i^{\text{eff}}$ enter with the same delayed-difference structure as the LISA laser phases. TDI therefore suppresses both the spacecraft and reference laser noise.

Finally, the upper-sideband-minus-carrier variable takes the form

$$\begin{aligned} r_{ij} &= \frac{\eta_{ij}^{\text{sb},+} - \eta_{ij}}{\nu_j^m} \\ &= \mathbf{D}_{ij} q_j - q_i + \frac{\mathbf{D}_{ij} m_j - m_i}{\nu_j^m} \\ &\quad + (1 - \mathbf{D}_{ij})\epsilon_i + \frac{b_{ij}^{\text{sb},+} - b_{ij}^c}{\nu_j^m}. \end{aligned}$$

The common GW phase cancels, leaving the clock-transfer signal together with modulation, board and readout noise.

The delay boards share an Rb reference. We write $\epsilon_i = \epsilon_{\text{com}} + \Delta\epsilon_i$ to separate shared timing from differences between boards. In the constant-frequency model, the shared term is absorbed by replacing the effective source phases with $\delta\phi_i^{\text{eff}} - \nu_{R,i}\epsilon_{\text{com}}$ and the clock errors with $q_i - \epsilon_{\text{com}}$. It therefore has the same residual delay operator as laser phase noise in this ideal model. This identifies the algebraic cancellation, not the residual amplitude in hardware; the latter depends on the input spectrum and errors in the delays used by TDI. Differential board timing remains a separate contribution.

Draft note: Document the distribution of the Rb reference to the ADCs, DACs and delay counters, and measure the differential timing between boards.

Tables I and II summarize the noise inputs. The optical readout calculation and the measured electronic and modulation spectra are discussed in Section IV.

The science-readout allocations are IRD2–IDS–04010 and IRD2–IDS–04580. Their ASDs rise as $\sqrt{1 + (f_k/f)^4}$, with $f_k = 2\text{ mHz}$ for the carrier and 0.7 mHz for the sideband. The optical-metrology comparison curve used later includes a different, total noise allocation.

TABLE I. LISA noise scales and the corresponding miniLISA inputs. Laser and readout allocations are plateau values; clock and measured component levels are quoted at 10 mHz. Predictions and experimental targets are not measured performance. The frequency dependence and measurement assumptions are given in Section IV.

Noise source	LISA	miniLISA	Treatment
Sources included in both			
Laser frequency noise	30 Hz/ $\sqrt{\text{Hz}}$ reference	Same reference adopted	Suppressed by TDI.
Spacecraft clock timing q_i	10.1 ps/ $\sqrt{\text{Hz}}$ model	Same level targeted	Sideband clock correction.
Modulation noise	No separate level assigned here	4.0×10^{-6} cycles/ $\sqrt{\text{Hz}}$ measured estimator	To be remeasured; contributes through clock correction.
Carrier readout	9 pm/ $\sqrt{\text{Hz}}$ allocation	0.0036 pm/ $\sqrt{\text{Hz}}$ calculation	Bright-beam input readout, including ideal ADC quantization.
Sideband readout	600 $\mu\text{rad}/\sqrt{\text{Hz}}$ allocation	0.0778 $\mu\text{rad}/\sqrt{\text{Hz}}$ calculation	Shared detector inputs have different propagation from independent link readouts.
ADC / phasemeter noise	Included in readout allocations	Electronic baseline in Table II	Not added twice in the combined curve.

TABLE II. Additional sources associated with the laboratory implementation. Levels at 10 mHz are illustrative input values. The electronic spectrum is measured. The boards share a timing reference; differential timing is assumed negligible in the plotted component sum.

Noise source	Level used	Treatment
Shared reference laser	Included in effective laser phases	Shared between all optical inputs; suppressed with laser noise by TDI.
Differential board timing $\Delta\epsilon_i$	No separate spectrum assigned	Common timing is suppressed by TDI and clock correction. Differential timing is assumed below the included contributions.
Electronic baseline	4.7×10^{-6} cycles/ $\sqrt{\text{Hz}}$ measured	Assigned to the carrier and sideband link readouts.
Delay-implementation residuals	Not quantified in this calculation	Development is ongoing; excluded from the component sum.

IV. NOISE CONTRIBUTIONS

We estimate the noise after TDI and clock correction by propagating measured electronic and modulation spectra together with a calculated detector contribution. For comparison, we use the optical-metrology and test-mass acceleration reference adopted by Ferguson et al. [9]. The following calculation identifies the relative contributions of these sources; it assumes matched physical and processing delays.

A. Modulation and Optical Readout Noise

Modulation noise and optical readout noise enter the measurement at different stages. Modulation noise is added while generating the sidebands used for clock transfer, as in LISA. Readout noise is added when measuring the carrier and sideband phases; it contributes to both the science measurement and clock correction.

We estimate modulation noise from half the phase dif-

ference between the upper and lower sidebands of a local optical beat. The common optical phase cancels, leaving modulation fluctuations and readout noise. We assign this measured spectrum to the direct modulation phase term, $m_i = \psi_i^{\text{mod}}$, with $\tau_i^{\text{mod}} = 0$. This assignment does not separate modulator noise from noise in the measurement itself.

Separately, the detector calculation gives carrier and sideband readout ASDs of approximately 3×10^{-9} and 1×10^{-8} cycles/ $\sqrt{\text{Hz}}$ for the assumed 4 mW per beam. These estimates lie well below the measured modulation and electronic spectra, indicating that optical readout should not limit this operating point. The relevant difference from LISA is the received optical power: miniLISA does not reproduce the weak received inter-spacecraft beam.

The modulation measurement already contains some readout noise, so adding the detector estimate separately can count a small contribution twice. The resulting sum is a component estimate, with this ambiguity well below the larger contributions.

B. Electronic Noise and Board Timing

The delay boards contribute additive electronic noise and timing fluctuations. We treat these separately: the measured baseline represents additive phase noise, while board timing couples through the input and output frequencies and the programmed delays.

We use the electronic delay-line baseline measured by Ferguson et al. [9] as an equivalent noise contribution for each carrier and sideband readout. The same ASD is assigned to every link and channel, with independent noise realizations. The baseline describes the measured chain without resolving its individual electronic components.

For timing, we write $\epsilon_i = \epsilon_{\text{com}} + \Delta\epsilon_i$, separating the shared reference from differences between boards. All appearances of the shared term are combined before calculating its noise power. In the constant-frequency model with matched delays, its residual has the same delay operator as laser phase noise. Thus the ideal TDI cancellation also applies to this term; it does not imply equal residual amplitudes for timing and laser noise. Residual power depends on the input spectrum, and a mismatch between physical and processing delays introduces additional leakage.

At the cancellation order retained here, the remaining board-timing contribution is

$$X_{2c}^\epsilon = \sum_i T_{\Delta\epsilon_i} \Delta\epsilon_i, \quad (19)$$

where $T_{\Delta\epsilon_i}$ is the complete delay response of the differential timing error on board i . The shared reference spectrum is not the spectrum of these differential terms.

We assume differential timing is below the electronic and modulation contributions and omit it from the plotted sum. Coupling of delay fluctuations to laser-frequency noise and errors in the implemented delays are also excluded; their size requires a measurement of the timing and delay residuals.

C. Propagation Through TDI

We follow the noise-propagation approach of Hartwig and Bayle [14]. First, each noise source is written in the carrier and sideband η measurements. We then form the clock-transfer variables and substitute the measurements into the corrected Michelson combination. Collecting all appearances of one source gives its complete delay polynomial and hence its transfer function. This includes both the physical link delays and the delays applied in processing.

Let $T_a(f)$ be the transfer function of source n_a , with PSD $S_a(f)$. For independent sources, the output PSD is

$$S_X(f) = \sum_a |T_a(f)|^2 S_a(f). \quad (20)$$

The order of these operations matters. A carrier readout reused in clock correction is the same noise source in both

occurrences, so its coefficients must be added before taking the squared magnitude. The same applies to a clock fluctuation shared between boards. Treating these occurrences as independent would remove the cancellations present in the measurement model.

For changing arms, the transfer functions also depend on the evaluation time. We calculate each response using the delays at that time, including the earlier times reached along a sequence of delays, and average the resulting noise powers over the chosen interval. This approximates the spectrum while the arms vary slowly. Appendix A specifies the delay model, averaging interval and checks on the calculation.

D. Combined Noise Contributions and Implications

Figure 4 shows the propagated electronic, modulation and detector contributions. The black curve is their quadrature sum, under the assumption that differential board timing is negligible. The dashed reference combines optical-metrology and test-mass acceleration noise, converted to phase at 1064 nm and propagated as independent equivalent link noises. Its input scales are $15 \text{ pm}/\sqrt{\text{Hz}}$ and $3 \text{ fm s}^{-2}/\sqrt{\text{Hz}}$, with the frequency dependence given in [2]. Test-mass noise is included only in this reference.

The detector contribution lies well below the electronic and modulation curves. Optical readout therefore does not set the calculated residual at the assumed powers. Among the contributions based on measured spectra, the electronic baseline is larger than the modulation estimate over most of the plotted band. Under the direct-phase-noise assumption, improving the modulation source alone would consequently leave the electronic contribution as the main limitation. This conclusion concerns the present operating point and the measured estimator; it does not establish the noise of the planned Moku modulation chain.

This conclusion assumes that differential board timing remains below the included contributions. The shared reference is combined coherently in the measurement model and does not enter as an independent board-noise floor.

The repeated minima at higher frequencies arise from the delayed differences in the TDI response. They occur in several component curves and in the propagated reference. A small ASD at one of these minima is therefore not evidence of a quieter instrument; a sensitivity comparison must also include the response to the injected GW signal. The dashed curve provides a common noise scale, but crossing below it does not by itself demonstrate LISA-like sensitivity or successful laser- and clock-noise suppression.

The calculation therefore indicates that noise cancellation can be studied at the assumed optical powers without a detector-noise limitation. The next useful improvements are to establish the differential board-timing con-

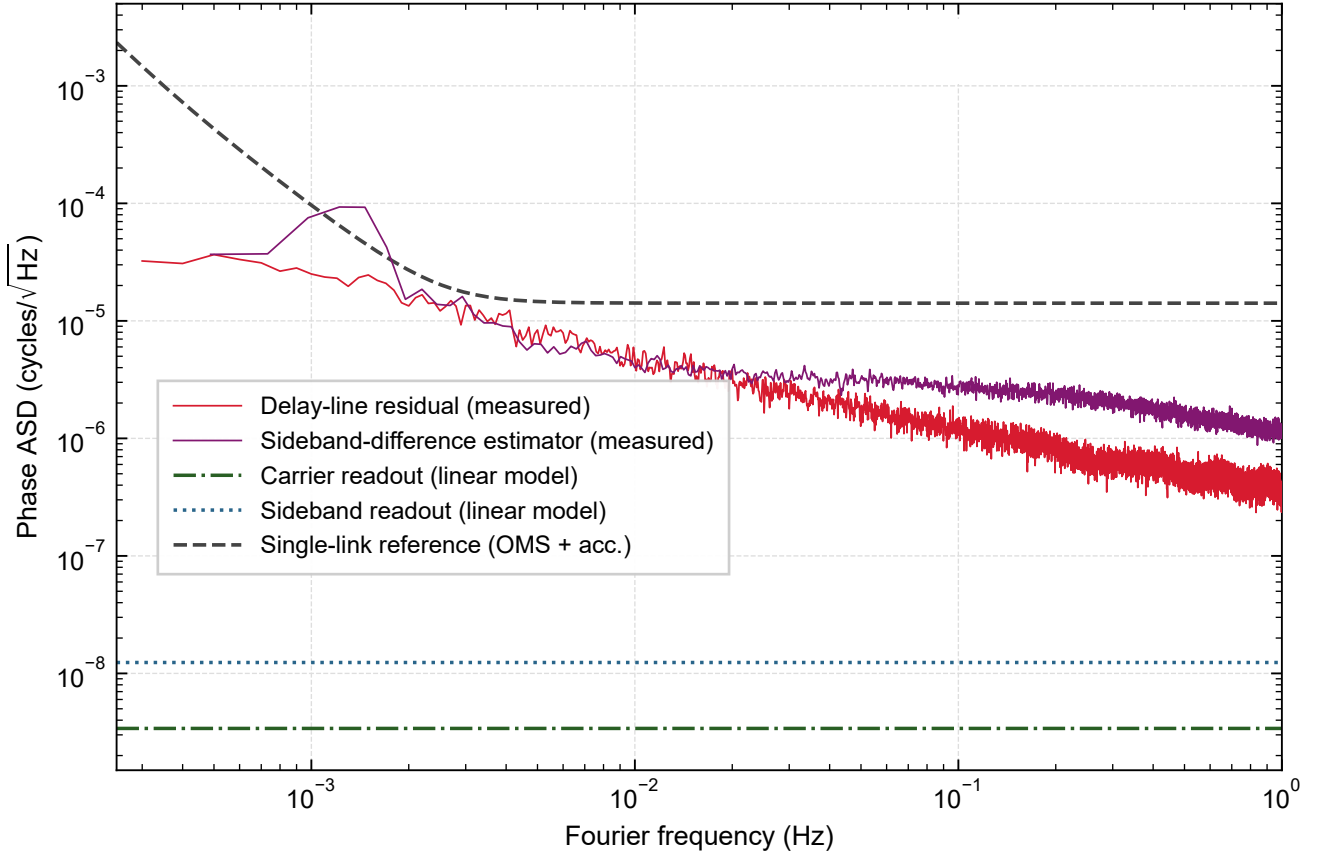


FIG. 3. Input noise spectra before TDI and clock correction. The electronic baseline and sideband-difference estimator are measured spectra; the carrier and sideband readout curves are order-of-magnitude calculations. The dashed curve is the single-link OMS-plus-acceleration reference.

tribution and reduce or characterize the electronic residual, while repeating the modulation measurement with the intended source and readout. Once delay-related residuals are included, the resulting budget can be tested against the assembled system and the recovery of a specified GW injection.

V. FUTURE EXTENSIONS

A. Changes to the Clock-Transfer and Locking Configuration

Changing the modulation frequency requires separating the measured estimator into modulation, timing and readout contributions, as discussed in Section II C. A GHz implementation would also require a compatible optical and RF frequency plan. Modulating the reference laser would allow corresponding optical sidebands to beat within the electronic input band.

A second optical source and modulator on each spacecraft, together with reference interferometers and back-

links, would reproduce the local $\xi/z/\eta$ reduction and the β clock couplings omitted here [12]. Distinct synthesis chains and a local sideband comparison would also allow tests of differential tone-generation noise.

B. Injection of Secondary Noise

Test-mass displacement can be emulated by adding the corresponding phase terms M_{ij} from Section II to the synthesized links. Each simulated test mass must use the same displacement realization wherever it occurs, including the delayed occurrence in the reverse link. Independent noise added to every link would not reproduce these correlations. For a specified acceleration spectrum, the displacement ASD is obtained by dividing by $(2\pi f)^2$ over the analysis band. This injection would test the secondary-noise transfer functions as well as primary-noise suppression.

Independent phase perturbations can also be added after link synthesis to test a prescribed LISA-like readout-noise model. Lower optical power would test weak-light

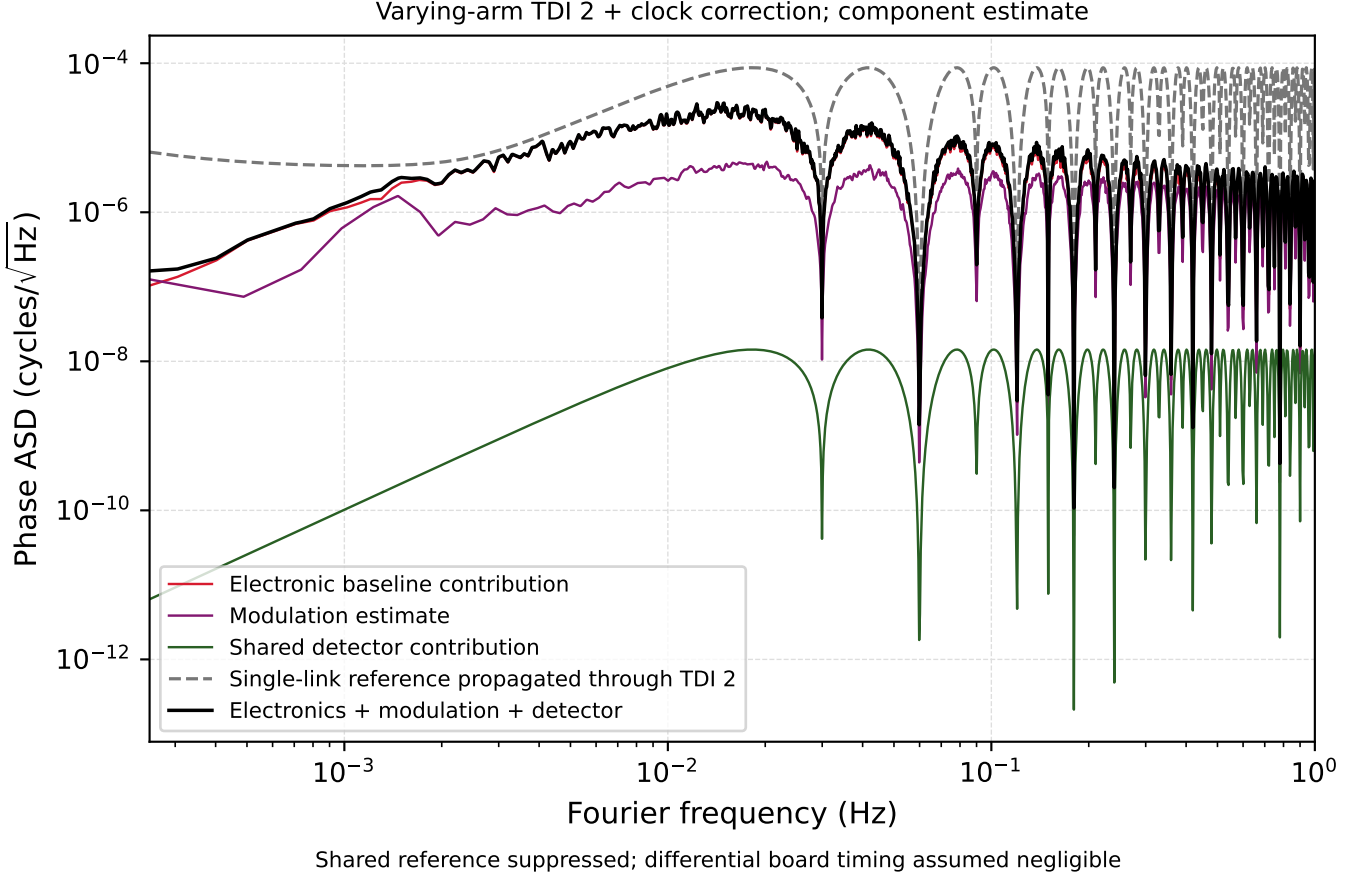


FIG. 4. Component phase-noise ASDs after second-generation Michelson TDI and clock correction, for the changing delays and constant beat frequencies in Appendix A. The black curve sums the electronic, modulation and detector terms. Shared board timing is suppressed by TDI and clock correction; differential timing is assumed negligible. The dashed curve is the propagated OMS-plus-acceleration reference. Delay-implementation residuals are excluded.

tracking and readout, but would not by itself remove the correlations introduced by splitting one detector output across boards.

VI. CONCLUSION

miniLISA combines optical laser signals, independent spacecraft clocks and electronic propagation delays to test TDI and clock correction in one measurement chain. The synthesized links reproduce the three-laser structure of the LISA measurements and a reduced sideband clock-transfer model. Adding prescribed GW phase responses allows the same experiment to test whether noise suppression preserves a known signal.

The component calculation indicates that the assumed laboratory optical powers keep detector readout below the measured electronic and modulation contributions. Electronics provide the larger measured contribution over most of the plotted band. The shared board-reference fluctuation is suppressed with laser noise, and differen-

tial timing is assumed below the included contributions. These results identify electronic noise and differential timing as priorities for characterisation, without assigning a complete sensitivity to the testbed.

The next steps are to repeat the modulation measurement with the intended source and readout, quantify differential board timing and delay errors, and compare the resulting budget with the assembled optical system. A specified GW injection will then test the recovered amplitude and phase alongside the achieved laser- and clock-noise suppression.

Appendix A: Parameters of the Component-Noise Calculation

We use the four links of the second-generation Michelson observable X_2 . Every delay is evaluated at the time reached after the preceding delays. For example,

$$\mathcal{D}_{12}\mathcal{D}_{21}x(t) = x(t - d_{12}(t) - d_{21}(t - d_{12}(t))). \quad (\text{A1})$$

The calculation retains this ordering rather than multiplying the first-generation result by a static filter.

The example uses $d_{12}(t) = d_{21}(t) = L + vt$ and $d_{13}(t) = d_{31}(t) = L - vt$, with $L = 8.339\text{ s}$ and $v = 5 \times 10^{-8}\text{ s/s}$. One arm grows while the other shrinks. This is a simple flexing-arm example, not an orbit reconstruction. The nominal electrical input frequencies are (15.0, 10.9, 21.6) MHz and the modulation frequencies are (35, 36, 34) MHz. The beat-frequency coefficients are held constant to isolate the effect of changing delays.

For this reduced measurement model, $\alpha_{j1} = -\alpha_{1j}$. The clock terms cancel in the corrected round trips

$$R_{1j}^c = \eta_{1j} + \mathcal{D}_{1j}\eta_{j1} - \alpha_{1j}r_{1j}, \quad j = 2, 3. \quad (\text{A2})$$

The corrected second-generation combination is then

$$X_{2c} = (1 - \mathcal{D}_{121} - \mathcal{D}_{12131} + \mathcal{D}_{1312121})R_{13}^c - (1 - \mathcal{D}_{131} - \mathcal{D}_{13121} + \mathcal{D}_{1213131})R_{12}^c. \quad (\text{A3})$$

Successive subscripts specify the ordered link path, for example $\mathcal{D}_{12131} = \mathcal{D}_{12}\mathcal{D}_{21}\mathcal{D}_{13}\mathcal{D}_{31}$. This is the phase-domain Michelson construction of [14]. It cancels laser noise through first order in the arm-length rates. The simple round-trip clock correction used here relies on constant, opposite forward and reverse beat frequencies; an orbit-dependent frequency plan requires the corresponding general clock correction.

For the spectral estimate, the interval is one day centred on $t = 0$. We evaluate the transfer functions at 25 equally spaced interval midpoints and average $|T_a(f, t)|^2$ for each source. This treats the slowly changing response as a sequence of local spectra; it does not calculate the exact spectrum of a time-warped stochastic record.

As a static check, setting all arm rates to zero and all one-way delays to L recovers the independent-link ASD gain $8|\sin(2\pi fL)\sin(4\pi fL)|$ of X_2 . A separate algebraic check shows that the surviving local-laser paths have equal delays through first order in arbitrary arm rates. Increasing the number of averaging times from 25 to 101 changes the retained component powers by less than 0.3% away from numerically vanishing responses. These are checks of the calculation, not measurements of hardware cancellation.

The current figure retains the electronic, modulation and detector curves from the saved vector output. Because the original measured input files are unavailable in this checkout, their quadrature sum was reconstructed by log-log interpolation of the saved curve vertices. This preserves the plotted spectra to vector-export precision; it is not a new evaluation of the raw measurements.

-
- [1] Benjamin P Abbott, Richard Abbott, Thomas D Abbott, Matthew R Abernathy, Fausto Acernese, Kendall Ackley, Carl Adams, Thomas Adams, Paolo Addesso, Rana X Adhikari, et al. Observation of gravitational waves from a binary black hole merger. *Physical review letters*, 116(6):061102, 2016.
 - [2] Monica Colpi, Karsten Danzmann, Martin Hewitson, Kelly Holley-Bockelmann, Philippe Jetzer, Gijs Nelemans, Antoine Petiteau, David Shoemaker, Carlos Sopuerta, Robin Stebbins, Nial Tanvir, Henry Ward, William Joseph Weber, Ira Thorpe, Anna Dauriskikh, Atul Deep, Ignacio Fernández Núñez, César García Marirrodriga, Martin Gehler, Jean-Philippe Hailain, Oliver Jennrich, Uwe Lammers, Jonan Larrañaga, Maike Lieser, Nora Lützgendorf, Waldemar Martens, Linda Mondin, Ana Piris Niño, Pau Amaro-Seoane, Manuel Arca Sedda, Pierre Auclair, Stanislav Babak, Quentin Baghi, Vishal Baibhav, Tessa Baker, Jean-Baptiste Bayle, Christopher Berry, Emanuele Berti, Guillaume Boileau, Matteo Bonetti, Richard Brito, Riccardo Buscicchio, Gianluca Calcagni, Pedro R. Capelo, Chiara Caprini, Andrea Caputo, Eleonora Castelli, Hsin-Yu Chen, Xian Chen, Alvin Chua, Gareth Davies, Andrea Derdzinski, Valerie Fiona Domcke, Daniela Doneva, Irna Dvorkin, Jose María Ezquiaga, Jonathan Gair, Zoltan Haiman, Ian Harry, Olaf Hartwig, Aurelien Hees, Anna Heffernan, Sascha Husa, David Izquierdo, Nikolaos Karmenis, Antoine Klein, Valeriya Korol, Natalia Korsakova, Thomas Kupfer, Danny Laghi, Astrid Lamberts, Shane Larson, Maude Le Jeune, Marek Lewicki, Tyson Littenberg, Eric Madsen, Alberto Mangiagli, Sylvain Marsat, Ivan Martin Vilchez, Andrea Maselli, Josh Mathews, Maarten van de Meent, Martina Muratore, Germano Nardini, Paolo Pani, Marco Peloso, Mauro Pieroni, Adam Pound, Hippolyte Quelquejay-Leclerc, Angelo Ricciardone, Elena Maria Rossi, Andrea Sartirana, Etienne Savalle, Laura Sberna, Alberto Sesana, Deirdre Shoemaker, Jacob Slutsky, Thomas Sotiriou, Lorenzo Speri, Martin Staab, Daniele Steer, Nicola Tamanini, Gianmassimo Tasinato, Jesus Torrado, Alejandro Torres-Orjuela, Alexandre Toubiana, Michele Vallisneri, Alberto Vecchio, Marta Volonteri, Kent Yagi, and Lorenz Zwick. LISA definition study report, 2024. Version Number: 1.
 - [3] Jean-Baptiste Bayle and Olaf Hartwig. Unified model for the LISA measurements and instrument simulations. *Physical Review D*, 107(8):083019, 2023.
 - [4] Gregory Weaver, Jeffrey Garstecki, and Samuel Reynolds. The performance of ultra-stable oscillators for the gravity recovery and interior laboratory (grail). In *Proceedings of the 42nd Annual Precise Time and Time*

- Interval Systems and Applications Meeting*, pages 369–380, 2010.
- [5] Massimo Tinto, F. B. Estabrook, and J. W. Armstrong. Time-delay interferometry for LISA. *Physical Review D*, 65(8):082003, 2002. Publisher: American Physical Society.
 - [6] Shawn J. Mitryk, Guido Mueller, and Josep Sanjuan. Hardware-based demonstration of time-delay interferometry and TDI-ranging with spacecraft motion effects. *Physical Review D*, 86(12):122006, 2012.
 - [7] Glenn de Vine, Brent Ware, Kirk McKenzie, Robert E. Spero, William M. Klipstein, and Daniel A. Shaddock. Experimental demonstration of time-delay interferometry for the laser interferometer space antenna. *Physical Review Letters*, 104:211103, 2010.
 - [8] M. Laporte, H. Halloin, E. Bréelle, C. Buy, P. Grüning, and P. Prat. Status of the LISA On Table experiment: a electro-optical simulator for LISA. *Journal of Physics: Conference Series*, 840:012014, 2017.
 - [9] Reid Ferguson, Olaf Hartwig, and Guido Mueller. Applying tdi to hardware-simulated data. *arXiv preprint arXiv:2505.22579*, 2025.
 - [10] Gerhard Heinzel, Juan José Esteban, Simon Barke, Markus Otto, Yan Wang, Antonio F Garcia, and Karsten Danzmann. Auxiliary functions of the **LISA** laser link: ranging, clock noise transfer and data communication. *Classical and Quantum Gravity*, 28(9):094008, 2011.
 - [11] Gerhard Heinzel, Javier Álvarez Vizoso, Miguel Dovale-Álvarez, and Karsten Wiesner. Frequency planning for LISA. *Physical Review D*, 110(4):042002, 2024.
 - [12] Markus Otto, Gerhard Heinzel, and Karsten Danzmann. Tdi and clock noise removal for the split interferometry configuration of lisa. *Classical and Quantum Gravity*, 29(20):205003, 2012.
 - [13] Massimo Tinto and Olaf Hartwig. Time-delay interferometry and clock-noise calibration. *Physical Review D*, 98(4):042003, 2018.
 - [14] Olaf Hartwig and Jean-Baptiste Bayle. Clock-jitter reduction in LISA time-delay interferometry combinations. *Physical Review D*, 103(12):123027, 2021. Publisher: American Physical Society.